

202504: Entering the R Console and Thinking Like a Risk Analyst

Theme: Introduction to R Language and Basic Statistics

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Programme: Nurturing Future Risk Managers and Actuaries: An AI-powered Journey for Gifted Secondary Students

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1 Mission Brief: Why This Tutorial Is Not Just “Learning Commands”

Opening Problem

An insurer receives a short sequence of six weekly claim amounts. A student opens a spreadsheet, types a few formulas, and quickly reports: “The average is stable, so the risk is under control.” Another student writes a short R script, checks the spread, draws a quick plot, and says: “The centre is not the whole story; volatility and shape matter.”

Driving question: If both students looked at the *same data*, why did one conclusion sound stronger? What extra power did the R workflow provide?

Your role today: You are not merely memorising syntax. You are learning how a risk analyst uses a programming language to move from **raw numbers** to **structured evidence**. In this tutorial, every command must be connected to an idea: representation, checking, interpretation, and communication.

Key Definition / Result

Big picture. R is useful not because it can “do calculations” faster than a human, but because it helps us build **repeatable, auditable, scalable** analyses. In risk work, this matters because we rarely trust a single number without asking:

- What was the input data structure?
- What function transformed it?

- Can we reproduce the result?
- What does the result mean in context?

2 Opening Problem: Why R Matters in Risk Management

Worked Example: A short argument worth unpacking

★ Core

A teacher says: “If you can already calculate with a calculator, then a programming language is optional.”

Task. Decide whether this claim is safe, sometimes safe, or misleading. Your answer must mention **at least two** of the following ideas:

- repeatability,
- checking many scenarios,
- visualisation,
- working with vectors or datasets,
- communication and collaboration.

Key Definition / Result

R as a risk tool. R is a free, open-source programming language widely used for modelling probabilities, simulating risks, analysing financial data, and communicating results through tables and graphs. In this course, we treat R not as a separate technical hobby, but as one of the main working languages of modern quantitative decision-making.

3 A. Learning with AI, but Verifying Like an Analyst

Key Definition / Result

AI tools can help students generate example code, explain error messages, and suggest alternative strategies. However, a useful principle for this programme is:

AI output $\neq$ final truth.

Every AI-generated answer must be checked by **running code, inspecting outputs**, and **asking whether the result makes sense in context**.

Suggested AI Prompt: Concept first, then code

★ Core

Explain what R is and why it is useful for risk management. Then give three examples of tasks in risk work for which R is more efficient than a basic calculator. For each example, say what the input is, what output is produced, and what interpretation a student should make.

Technology Check: AI-assisted study protocol

★★ Insight

When using an AI assistant, make students explicitly separate these four stages:

T1: Ask for explanation. What does the command do conceptually?

T2: Ask for code. What is the R syntax?

T3: Run the code. What actually appears in the console or plot window?

T4: Audit the answer. Does the output match the question? Can a small hand check confirm it?

4 B. Access Routes: Local Installation and Cloud Use

Key Definition / Result

Students can work with R in two broad ways:

- **Local installation:** install R from the official website and run it on your own computer.
- **Browser-based access:** use a cloud workspace such as Posit Cloud to avoid local installation in the early learning phase.

Each pathway is valid, but the second often lowers the barrier for beginners because it removes the software-installation burden.

Worked Example: Choosing an access route

★ Core

Question. A student wants to begin immediately and is worried about installation issues. Which route is more suitable today, and why?

Micro-steps.

- (1) Local installation is powerful and permanent, but it requires downloading and setting up software correctly.
- (2) A browser-based platform gets the student into an RStudio-style environment quickly.
- (3) For a first contact lesson, the main goal is not operating-system troubleshooting; it is learning input, output, and interpretation.

Technology Check: Installation pathway snapshot

★ Core

Local installation pathway.

- Visit <http://www.r-project.org/>.

- Click **CRAN** (Comprehensive R Archive Network).
- Choose a mirror and download the correct version for your operating system.
- Follow the installation instructions, then open the R console.

Browser-based pathway.

- Visit <http://posit.cloud/>.
- Sign up and choose the cloud free option.
- Create a new project to open a browser-based RStudio workspace.

Suggested AI Prompt: Installation + access explanation

★ Core

Show me how to install R on my computer and how to access the R console. Explain what CRAN is, why mirrors exist, and how a beginner could choose between local installation and a browser-based cloud workspace.

5 C. First Contact with the Console: Input, Output, and Syntax

Key Definition / Result

Console mindset. The R console is a place where you type *commands* and receive *outputs*. The most important beginner habit is to read output carefully rather than only celebrating that “something appeared.”

Worked Example: The smallest possible experiment

★ Core

Type the following into the console:

```
1 + 1
3 * 2
(1:3) + 2
```

Prompt for thought. What changes when you move from a single arithmetic

statement to a vector operation?

Common Mistake: Treating the console like a word processor

Students sometimes type explanatory English text directly after code on the same line and expect R to “understand the intention.” It does not. R reads syntax, not casual prose. If you want explanation, use a comment with the symbol #.

6 D. Functions: What They Are and How to Read Them

Key Definition / Result

A function in R takes an input and returns an output. The general form is:

```
function_name(data_object).
```

The command name tells R what operation to perform; the parentheses tell R which object or values the operation should act on.

Worked Example: Reading `mean(my_data)`

★ Core

Question. What are the two most important pieces of information conveyed by `mean(my_data)`?

Worked Example: Why functions matter in statistics**★★ Insight**

Suppose a student calculates a mean by hand once and says, “I do not need a function.” Why is that too narrow a view?

7 E. Help System: Learning to Ask the Software

Key Definition / Result

R has a built-in help system. Two common ways to query it are:

`help(function_name)` or `?function_name`

These commands are important because independent learners should not rely only on memory or internet searching.

Worked Example: When you forget a function’s purpose**★ Core**

A student remembers that `sd()` has something to do with variation, but cannot recall the details.

Task. Write the exact command that opens the help page and then explain why knowing how to consult documentation is a professional habit.

Suggested AI Prompt: Help system as a learning habit★ **Core**

Explain how to get help in R using the `help()` function or the `?` shortcut. Give examples with `mean`, `sum`, and `plot`. For each one, say what kind of information a student should look for in the help page before using the function.

8 F. Comments: Code That Speaks to Humans

Key Definition / Result

Comments in R begin with `#`. They are not executed as code. Their purpose is to explain intention, document logic, record assumptions, and improve readability for both the author and collaborators.

Worked Example: Detecting why one line fails and one line works

★

Core

Study the following interaction:

```
# It is an example.  
1 + 1 I am a comment!  
# Error: unexpected symbol ...  
3 * 2 # I am a comment!  
# Bye!
```

Question. Why does the second arithmetic line work while the first fails?

Common Mistake: Using comments only as decoration

A weak habit is to write comments that merely repeat the code, such as “`x <- 5 # x becomes 5`.” A stronger habit is to explain purpose and meaning, for

example: “`x <- 5 # baseline annual premium in thousands of dollars.`”

9 G. Core Command Families: From Arithmetic to Data Structure

9.1 Assignment, arithmetic, and logical checks

Key Definition / Result

The first cluster of commands students must control includes:

- assignment: `<-` or `=`
- arithmetic: `+`, `-`, `*`, `/`, `^` or `**`
- logical checks: `==`, `!=`, `<`, `<=`, `>`, `>=`

These commands already introduce an important idea: some outputs are **numbers**, while others are **truth values** such as `TRUE` or `FALSE`.

Worked Example: Assignment is not just storage

★ Core

Consider:

```
202504 <- 202504
```

What has changed after this command is executed?

Worked Example: Logical checks as filters for decisions

★★ Insight

Interpret the output of:

```
(1:3) >= 2
```

9.2 Vectors, sequences, repetition, and extraction

Key Definition / Result

A vector is one of the most important basic structures in R. Useful constructors and operations include:

- `c()` for combining values,
- `:` and `seq()` for generating regular sequences,
- `rep()` for repetition,
- `[]` for extraction or selection.

Without vectors, statistical thinking in R remains very limited.

Worked Example: Why vectors matter for risk

★ Core

A single claim amount tells you very little. A vector of claim amounts opens the door to patterns. Explain why `c(1200, 950, 1300, 870, 2200)` is more valuable analytically than a lone number.

Worked Example: Extraction as a targeted question

★★ Insight

Consider the vector `c(1,5,0,9)`.

- `[2:3]` asks for positions 2 and 3.
- `[-2]` asks for everything except position 2.
- `[c(F,F,T,T)]` asks for entries whose mask is TRUE.

10 H. Mathematical Functions: Transforming Values with Meaning

Key Definition / Result

Important mathematical functions include `abs()`, `sqrt()`, `round()`, `exp()`, `log()`, `max()`, `min()`, and `sort()`.

Worked Example: Absolute value in a risk context

★ Core

Suppose forecast error can be positive or negative. Why might `abs(error)` be more useful than the signed error itself in some situations?

Worked Example: `exp()` and `log()` as structural opposites

★★ Insight

Explain the relationship between `exp()` and `log()` in one conceptual paragraph.

11 I. Statistical Functions: From Raw Data to Summary Judgement

Key Definition / Result

Core statistical functions in this lesson:

- `sum()` : total amount
- `mean()` : arithmetic centre
- `var()` : sample variance
- `sd()` : sample standard deviation

These functions are the first step from “a list of numbers” toward “a statistical description of behaviour.”

Worked Example: Mean and variation tell different stories ★ Core

Two portfolios can share the same mean weekly return but have very different variability. Why does this already justify learning `var()` and `sd()` as soon as `mean()`?

Worked Example: A tiny claims dataset ★★ Insight

Let

```
claims <- c(1, 4, 2, 5, 8)
```

Explain what each of the following tells us conceptually:

```
sum(claims)
mean(claims)
var(claims)
```

```
sd(claims)
```

12 J. Graphing Functions: Seeing Structure, Not Just Producing Pictures

Key Definition / Result

Useful basic graphing functions include `plot()`, `hist()`, `boxplot()`, `barplot()`, `pie()`, `points()`, `lines()`, and `curve()`.

Worked Example: Choosing the right plot

★ Core

Match each question with a suitable graphing idea:

- (a) What does the distribution of a single variable look like?
- (b) How do two variables move together?
- (c) How can I add a second fitted or theoretical curve onto an existing graph?

Common Mistake: Confusing graph type with graph purpose

Students sometimes choose a graph because “it looks nice.” Instead, ask: What pattern am I trying to reveal? For example, a pie chart gives category

proportions, while a histogram shows distribution shape. The correct choice depends on the question, not on decoration.

13 K. Worked Example: Sine and Cosine as a Graphing Laboratory

Worked Example: Reading and extending a short plotting script

★

Core

Study the following code:

```
# Generating a sequence of values for x from 0 to 2*pi with a length of
  100
x <- seq(0, 2*pi, length.out = 100)
# Computing the sine and cosine of each value of x
y1 <- sin(x)
y2 <- cos(x)
# Plotting the sine function with a line plot
plot(x, y1, type = "l", main = "Sine and Cosine Functions")
# Adding the cosine function to the same plot with a red line
curve(cos(x), from = 0, to = 2*pi, add = TRUE, col = "red")
```

Micro-steps for interpretation.

- Step 1: Input design.** The vector x is not random; it is a carefully chosen grid of 100 values from 0 to 2π .
- Step 2: Transformation.** $\sin(x)$ and $\cos(x)$ apply a function to every element of that grid.
- Step 3: First display.** `plot(..., type = "l")` displays the sine function as a line.
- Step 4: Overlay.** `curve(..., add = TRUE)` places the cosine function on the same axes for comparison.
- Step 5: Interpretive value.** One graph now shows similarity, phase shift, and periodic structure.

Technology Check: Modify the code, not merely run it**** Insight**

After you reproduce the graph, make one change at a time:

- add axis labels,
- change the line colours,
- explicitly plot both `y1` and `y2` using `lines()`,
- increase or decrease the `length.out` value,
- explain whether fewer grid points make the curve look rougher.

14 L. Investigation: Build a Mini Risk Data Workflow

Investigation / Activity: From five claim amounts to a small analytic story**** Insight**

Define the following vector in R:

```
claims <- c(1200, 950, 1300, 870, 2200)
```

Complete the following workflow.

- I1:** Compute the total, mean, variance, and standard deviation.
- I2:** Sort the claims and identify the largest one.
- I3:** Produce a simple histogram or barplot.
- I4:** Write three sentences interpreting the data in words rather than just reporting outputs.
- I5:** State one limitation of drawing conclusions from a dataset this small.

Suggested AI Prompt: Use AI to deepen interpretation rather than to bypass it ★★ **Insight**

I have a small R dataset of claim amounts. Suggest a short analysis plan using `sum()`, `mean()`, `var()`, `sd()`, and one graph. Then explain how each result should be interpreted in an insurance or risk context. Do not only give code; explain the reasoning behind each step.

15 M. Concept Map: What Students Should Leave With Today

Key Definition / Result

By the end of this tutorial, students should understand the following chain:

raw values → named objects → vectors → functions → summaries / plots →
interpretation in context

This chain is the real learning target. The command list only matters because it supports this progression.

16 Review Set 1A: Foundational Practice

Worked Example: Short review questions

★ **Core**

Answer the following.

- A1:** In one or two sentences, explain why R is useful in risk management.
- A2:** What is the difference between local installation and a cloud workspace?
- A3:** Write one command using `help()` and one command using `?`.
- A4:** Why do comments matter?
- A5:** What does `c(1,3,5,7)` create?
- A6:** What is the difference between `sum()` and `mean()`?

A7: Which graphing command would you start with for a single numerical variable?

A8: Explain what $(1:3) > 2$ is doing conceptually.

17 Review Set 1B: Deeper Thinking and Transfer

Worked Example: More demanding questions

★★ Insight

B1: A student says: “The mean is enough to describe a risk dataset.” Build a counterargument using the idea of variability.

B2: Why is `#` a small symbol with large professional importance?

B3: Compare `seq()` and `:`. When might one be more flexible than the other?

B4: Explain why logical outputs such as `TRUE` and `FALSE` are useful in real workflows rather than being merely “computer language.”

B5: A student produces a beautiful graph but cannot explain what question it answers. What is structurally missing from the work?

B6: The code runs, but the student has no comment lines, no labels, and no interpretation. Why is this still incomplete analysis?

18 Conclusion: The One-Sentence Rule

Worked Example: Exit ticket

★ Core

Complete this sentence in a mathematically meaningful way:

*“R becomes powerful in risk management when it lets us move from
_____.”*

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Key Definition / Result

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